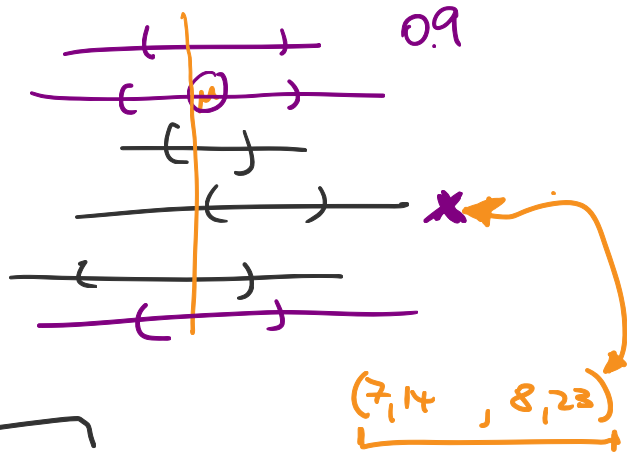
$\boxed{n=25}$ Enf

$X_{11} = \quad X_{12} \quad \cdots \quad X_{1,25}$

$\vdots$

$X_{B1} \quad X_{B2} \quad X_{B,25}.$

$\boxed{n=100}$ Sanos.



$IC_{0.95}(\mu_j)$ $\boxed{1 \le j \le B}$

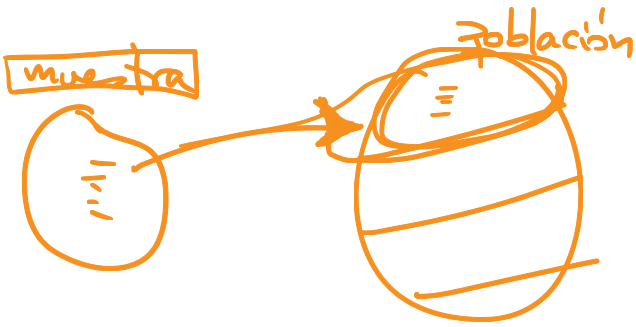
$P(\mu_1 \in I_1) = 0{,}95$

$P(\mu_2 \in I_2) = 0{,}95.$

$$\prod_{i=1}^{2} P(\mu_i \in I_i) \underset{\text{indep.}}{=} P(\{\mu_1 \in I_1\} \cap \{\mu_2 \in I_2\}) \overset{?}{=} 0.95$$

complementando

$$= 1 - \underbrace{P\left((\mu_1 \in I_1)^c \cup (\mu_2 \in I_2^c)\right)}_{\text{Desig. de Bonferroni}}$$



Ej 1 _ Lista 10.

$$f_{X_1}(x;\theta) = \frac{2}{\theta^2} x \, I_{(0,\theta]}(x) \qquad (\theta > 0)$$

a) Nos piden $\hat{\theta}_{MV}$. basado en $X_1, \dots, X_n$

son iguales

$$L(\theta) = f_{X_1 \dots X_n}(x_1, \dots, x_n; \theta) \underset{X_1,\dots,X_n \text{ indep.}}{=} \prod_{i=1}^{n} f_{X_i}(x_i;\theta) = \prod_{i=1}^{n} \left\{ \frac{2}{\theta^2} x_i \, I_{(0,\theta]}(x_i) \right\}$$

$$= \frac{2^n}{\theta^{2n}} \left(\prod_{i=1}^{n} x_i \right) \boxed{\prod_{i=1}^{n} I_{(0,\theta]}(x_i)}$$

nos resta escribir este producto como función de θ.

¿cuánto vale el producto de indicadoras?

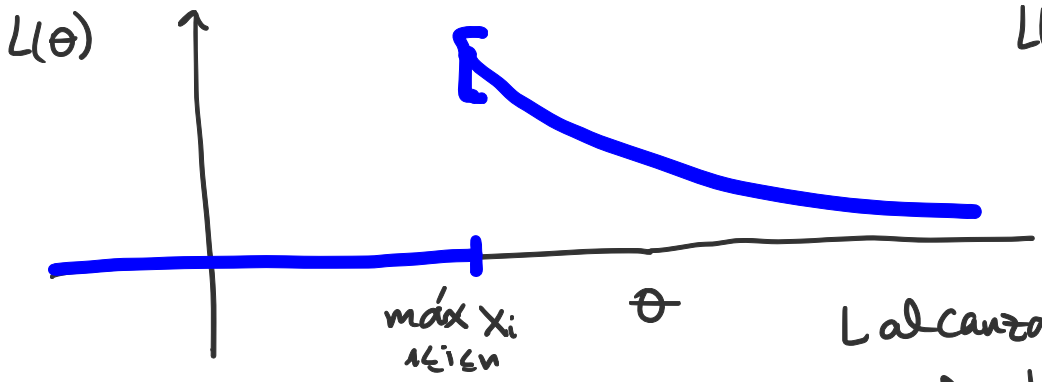
$$\prod_{i=1}^{n} I_{(0,\theta]}(x_i) = \begin{cases} 1 & \text{si y sólo si } 0 < x_i \leq \theta \ \forall\, 1 \leq i \leq n \\ 0 & \text{sino} \end{cases}$$

$$0 < x_i \leq \theta \iff \theta \geq x_i \ \forall i = 1, \dots, n \iff \theta \geq \max_{1 \leq i \leq n} \{x_i\}$$

$$\iff I_{[\max_{1 \leq i \leq n}\{x_i\}, +\infty)}(\theta) = 1.$$

Luego $L(\theta) = \dfrac{2^n}{\theta^{n2}} \left(\prod_{i=1}^{n} x_i \right) \cdot I_{[\max_{1 \leq i \leq n}\{x_i\}, +\infty)}(\theta)\, I_{(0,+\infty)}(\min x_i)$

$\hat{\theta}_{MV}$ será el valor de θ en el cual L alcanza el máximo



$$L(\theta) = \begin{cases} 0 & \text{si } \theta < \max X_i \\ a(\theta) & \text{si } \theta \geq \max X_i \end{cases}$$

es decreciente.

L alcanza su máximo cuando θ vale $\max_{1 \leq i \leq n}\{X_i\}$

$$\Rightarrow \hat{\theta}_{MV} = \max_{1 \leq i \leq n} X_i$$

b) ¿será insesgado? Necesitamos $E_\theta\left(\max_{i=1,\dots,n} X_i\right)$ Para calcularla

- o bien usamos propiedades
- o bien hallamos $f_{\hat{\theta}}$

$$F_{\hat{\theta}}(t) = P\left(\max_{1 \leq i \leq n} X_i \leq t\right) = P\left(\bigcap_{i=1}^{n} \{X_i \leq t\}\right) \underset{\text{indep}}{=} \prod_{i=1}^{n} \underbrace{P(X_i \leq t)}_{F_{X_i}(t)}.$$

$$\underset{\text{id}}{=} \left[F_{X_1}(t)\right]^n$$

$$\Rightarrow f_{\hat{\theta}}(t) = n \left[F_{X_1}(t)\right]^{n-1} \cdot f_{X_1}(t). \qquad (1)$$

$$F_{X_1}(t) = \int_{-\infty}^{t} f_{X_1}(s)\,ds = \int_{-\infty}^{t} \frac{2s}{\theta^2} I_{(0,\theta]}(s)\,ds = \begin{cases} 0 & \text{si } t \leq 0. \\ \int_0^t \frac{2s}{\theta^2}\,ds & \text{si } 0 < t < \theta \\ \int_0^\theta \frac{2s}{\theta^2}\,ds & t \geq \theta \end{cases}$$

$$\boxed{F_{X_1}(t) = \begin{cases} 0 & \text{si } t \leq 0 \\ t^2/\theta^2 & \text{si } 0 < t \leq \theta \\ 1 & \text{si } t > \theta. \end{cases}}$$

$$\int_0^t \frac{2s}{\theta^2}\,ds = \left.\frac{2s^2}{2\theta^2}\right|_{s=0}^{s=t} = \frac{t^2}{\theta^2}$$

Luego

$$f_{\hat{\theta}}(t) = n\underbrace{\left[\left(t^2/\theta^2\right)^{n-1} I_{(0,\theta]}(t) + 1^{n-1} I_{(\theta,+\infty)}(t)\right]}_{[F_{X_1}(t)]^{n-1}} \underbrace{\frac{2t}{\theta^2} I_{(0,\theta]}(t)}_{f_{X_1}(t)}.$$

$$= \frac{2n\,t^{2n-1}}{\theta^{2n-2+2}} I_{(0,\theta]}(t). = \frac{2n\,t^{2n-1}}{\theta^{2n}} I_{(0,\theta]}(t).$$

$$\boxed{E_\theta(\hat{\theta})} = E(\text{máx}\,X_i) = \int_{-\infty}^{+\infty} t\, f_{\hat{\theta}}(t)\,dt = \int_{-\infty}^{+\infty} \frac{2n}{\theta^{2n}} t^{2n} I_{(0,\theta]}(t)\,dt$$

$$= \frac{2n}{\theta^{2n}} \left.\left(\frac{t^{2n+1}}{2n+1}\right)\right|_{t=0}^{t=\theta} = \frac{2n}{(2n+1)\theta^{2n}}\theta^{2n+1} = \boxed{\frac{2n\theta}{2n+1}} \neq \theta$$

¿Es insesgado $\hat{\theta}$? Lo sería si $E(\hat{\theta}_n) = \theta$

No, no es insesgado.

¿Es asintóticamente insesgado? Lo sería si:

$$E(\hat{\theta}_n) \xrightarrow{n\to\infty} \theta.$$

$$\left(\frac{2n}{2n+1}\right)\theta \xrightarrow[n\to\infty]{} \theta. \quad \text{Lo es. } \checkmark$$

c) Nos piden el $ECM(\hat{\theta})$ $\quad ECM(\hat{\theta},\theta)$

$$ECM(\hat{\theta},\theta) \underset{\substack{\uparrow \\ \text{ej5 P10}}}{=} Var(\hat{\theta}) + \underbrace{\left(E(\hat{\theta}) - \theta\right)}_{\text{sesgo.}}^2 \qquad \text{cuando } X_i \sim F(\cdot;\theta)$$

$$Var(\hat{\theta}) = E(\hat{\theta}^2) - E(\hat{\theta})^2$$

$$E(\hat{\theta}^2) = \int_{-\infty}^{+\infty} t^2 \underbrace{f_{\hat{\theta}}(t)}_{\frac{2n}{\theta^{2n}} t^{2n-1} I_{(0,\theta]}(t).} dt = \int_0^{\theta} \frac{2n}{\theta^{2n}} t^{2n+1} dt =$$

$$= \frac{2n}{\theta^{2n}} \left. \frac{t^{2n+2}}{2n+2} \right|_{t=0}^{t=\theta} = \frac{2n}{\theta^{2n}} \frac{\theta^{2n+2}}{2n+2} = \frac{2n}{2n+2}\theta^2$$

$$ECM(\hat{\theta},\theta) = Var(\hat{\theta}) + \left(E(\hat{\theta}) - \theta\right)^2$$

$$= \overbrace{\frac{2n}{2n+2}\theta^2}^{E(\hat{\theta}^2)} - \overbrace{\left(\frac{2n}{2n+1}\theta\right)^2}^{[E(\hat{\theta})]^2} + \left(\frac{2n}{2n+1}\theta - \theta\right)^2$$

$$= \frac{\theta^2\left[2n(2n+1)^2 - (2n)^2 2(n+1)\right]}{(2n+2)(2n+1)^2} + \left(\frac{2n\theta - (2n+1)\theta}{2n+1}\right)^2$$

$$2\,2n(n+1) = 4n^2 + 4n$$

$$= \frac{\theta^2\, 2n\left[\cancel{4n^2} + \cancel{4n} + 1 - \cancel{4n^2} - \cancel{4n}\right]}{2^2(n+1)^2(2n+1)^2} + \frac{\theta^2}{(2n+1)^2}$$

$$= \frac{\theta^2 \cancel{2n}}{\underset{2}{\cancel{4}}(n+1)^2(2n+1)^2} + \frac{\theta^2}{(2n+1)^2} = \frac{\theta^2}{(2n+1)^2\, 2}\overbrace{\left[\frac{1 + 2(n+1)^2}{(n+1)^2}\right]}^{2n^2+4n+2}$$

$$= \frac{2\theta^2(n^2+2n+1)}{2(2n+1)^2} \qquad \text{chequear.}$$

d) ¿Será $\hat{\theta}_n$ consistente? débilmente.

Calcular, $\varepsilon > 0$

$\hookrightarrow P(|\hat{\theta} - \theta| > \varepsilon) \xrightarrow{?} 0 \quad \forall \varepsilon$.

Como conocemos $f_{\hat{\theta}}$ $(0 < \varepsilon < \theta)$

$$P(|\hat{\theta} - \theta| > \varepsilon) \underset{\text{verificar}}{=} \left(\frac{\theta - \varepsilon}{\theta}\right)^{2n} = \left(1 - \frac{\varepsilon}{\theta}\right)^{2n} \xrightarrow[n \to \infty]{} 0$$

$0 < 1 - \frac{\varepsilon}{\theta} < 1$. Sí.

Otra forma.

Prop: $\hat{\theta}_n$ es un estimador de θ, tq:

a) $E(\hat{\theta}_n) \to \theta \quad n \to \infty$.

b) $Var(\hat{\theta}_n) \to 0$.

Entonces $\hat{\theta}_n$ es consistente. (débilmente)

PROBARLO

Usar b.